

# LS10 Guided Wave Radar Level Transmitter

SKU / Model Number: LS10-GWR-4M

Manufacturer: LevelPro Instruments

Category: Sensors | ETIM Code: EC001356

## PRODUCT OVERVIEW

The LS10 guided wave radar level transmitter uses Time Domain Reflectometry (TDR) technology for continuous level measurement. Suitable for liquids, slurries, and granular solids with foam or vapor presence.

## TECHNICAL SPECIFICATIONS

|                             |                         |
|-----------------------------|-------------------------|
| Measurement Principle       | Guided Wave Radar (TDR) |
| Measurement Range           | 0.1 to 4 m              |
| Accuracy                    | +/- 2 mm                |
| Resolution                  | 1 mm                    |
| Dielectric Constant Minimum | > 1.4                   |
| Probe Type                  | Single rod / Coaxial    |
| Update Rate                 | 1 Hz                    |

## ELECTRICAL RATINGS

|                   |                  |
|-------------------|------------------|
| Supply Voltage    | 12 to 36 V DC    |
| Output            | 4-20 mA + HART 7 |
| Power Consumption | < 1 W            |
| Load Resistance   | Max 600 ohm      |
| Display           | 4-digit LCD      |

**DIMENSIONS & MECHANICAL**

|                    |                         |
|--------------------|-------------------------|
| Probe Length       | Up to 4000 mm           |
| Process Connection | G 1-1/2 / 1-1/2 NPT     |
| Housing            | 66 x 100 mm             |
| Cable Entry        | M20x1.5                 |
| Weight             | 1.8 kg (with 1 m probe) |

**OPERATING CONDITIONS**

|                     |                  |
|---------------------|------------------|
| Process Temperature | -40 to 200 deg C |
| Process Pressure    | -1 to 40 bar     |
| Ambient Temperature | -40 to 70 deg C  |
| IP Rating           | IP66 / IP67      |
| Hazardous Area      | Zone 0 / Div 1   |

**MATERIALS OF CONSTRUCTION**

|              |                      |
|--------------|----------------------|
| Wetted Parts | 316L Stainless Steel |
| Probe Rod    | 316L Stainless Steel |
| Process Seal | PEEK / FKM           |
| Housing      | 316L Stainless Steel |
| Gasket       | PTFE                 |

**PERFORMANCE DATA / SELECTION TABLE**

| Medium | Dielectric | Min Range | Accuracy |
|--------|------------|-----------|----------|
| Water  | 80         | 0.1 m     | +/- 2 mm |
| Diesel | 2.1        | 0.3 m     | +/- 3 mm |
| LPG    | 1.6        | 0.5 m     | +/- 5 mm |
| Grain  | 3-5        | 0.3 m     | +/- 5 mm |

**CERTIFICATIONS & APPROVALS**

CE | ATEX | IECEx | SIL 2 | FM / CSA | RoHS

*Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.*

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